

SONiX 32-Bit Cortex-M0 Micro-Controller

Clock Switch Example Code

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AMENDMENT HISTORY

Version	Date	Description
1.0	2024/12/02	1. First version.

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1 OVERVIEW

This document provides our customers the C program examples of switching system clock.

2 FEATURE

- Switch System clock (HCLK) to IHRC, ILRC
- Enable/Disable Clockout (= IHRC, ILRC)
- Support Prescaler
- Use ARM-compiler V6.12

3 HW

- The Starter kit board of vendor desired MCU.

4 EXAMPLE CODE

1. Set SWITCH_CASE to desired system clock source in main.c, the related codes are provided in main function.

```
25 /*____DECLARATIONS____*/
26 void PFPA_Init(void);
27
28
29 /*____DEFINITIONS____*/
30 #define SWITCH_CASE    SYS0_CLK_IHRC    //SYS0_CLK_IHRC, SYS0_CLK_ILRC, SYS0_CLK_EHS, SYS0_CLK_ELS, SYS0_CLK_PLL
31
32
33 /*____MACROS____*/
34
35
```

2. If we want to enable Clockout function, call SYS1_EnableClockout and pass the desired arguments in main function.

For example, clockout HCLK to CLOCKOUT pin, and the output frequency is HLCK/2.

```
83
84 //Clockout (Optional)
85 SYS1_EnableClockout(SYS1_CLOCKOUT_HCLK, SYS1_CLOCKOUT_DIV2);
86
87 while (1);
88
89 }
```

★ Note: Except SN32F700/100/760/240, the FW SHALL call SystemCoreClockUpdate function with exact core frequency.

```
#if (SWITCH_CASE == SYS0_CLK_IHRC)
SYS0_EnableIHRC();
SYS0_SystemClockSwitch(SYS0_SYSCLK_IHRC, SYS0_AHBPRE_DIV1);
SYS0_SystemCoreClockUpdate(12000000); //Note: Must be called with Exact HCLK frequency, Do NOT remove!!!
#endif
#if (SWITCH_CASE == SYS0_CLK_ILRC)
SYS0_SystemClockSwitch(SYS0_SYSCLK_ILRC, SYS0_AHBPRE_DIV2);
SYS0_SystemCoreClockUpdate(8000); //Note: Must be called with Exact HCLK frequency, Do NOT remove!!!
#endif
#if (SWITCH_CASE == SYS0_CLK_EHS)
SYS0_EnableEHSXtal(16);
SYS0_SystemClockSwitch(SYS0_CLK_EHS, SYS0_AHBPRE_DIV1);
SYS0_SystemCoreClockUpdate(16000000); //Note: Must be called with Exact HCLK frequency, Do NOT remove!!!
#endif
#if (SWITCH_CASE == SYS0_CLK_ELS)
SYS0_EnableELSXtal();
SYS0_SystemClockSwitch(SYS0_CLK_ELS, SYS0_AHBPRE_DIV1);
SYS0_SystemCoreClockUpdate(32768); //Note: Must be called with Exact HCLK frequency, Do NOT remove!!!
#endif
#if (SWITCH_CASE == SYS0_CLK_PLL)
//Example: EHS=16MHz ---> PLL=48MHz
//Note: User shall call SYS0_EnableEHSXtal or SYS0_EnableIHRC to make sure
// the clock source of PLL is enabled and ready.
SYS0_EnableEHSXtal(16);
SYS0_EnablePLL(SYS0_PLL_M6, SYS0_PLL_P2, SYS0_PLL_CLOCK_EHS);
SYS0_SystemClockSwitch(SYS0_CLK_PLL, SYS0_AHBPRE_DIV1);
SYS0_SystemCoreClockUpdate(48000000); //Note: Must be called with Exact HCLK frequency, Do NOT remove!!!
#endif
```

```
int main (void)
{
    //Switch clock & Enable Clockout on demand
    SYS0_SystemClockSwitch(SYS0_CLK_ELS, SYS0_AHBPRE_DIV1);
    SYS0_SystemCoreClockUpdate(系統頻率值);

    //Clockout (Optional)
    SYS1_EnableClockout(SYS1_CLOCKOUT_HCLK, SYS1_CLOCKOUT_DIV2);

    .
    .
    User code
    .
    .
}
```

5 RESULT

The clockout waveform could be measured from CLOCKOUT pin. Due to the IO driving ability and loading limits, the output frequency is suggested to be less than 30MHz.

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